Verifiable Private Governance on Ethereum

A Governance Primitive for Encrypted Decisions and On-Chain Proofs

Ethereum Foundation PhD Fellowship 2026 — Proposal
RFP: Programmable Institutional Design and Verifiable Governance

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Project Abstract

Ethereum already supports highly programmable governance, but private collective decisions remain difficult to implement without introducing privileged workflow actors or sacrificing verifiability and efficiency. This project studies that gap through a concrete governance primitive: securely binding an encrypted decision to the succinct proof that authorizes it when the two live in incompatible algebraic domains. The flagship case is threshold-Paillier encryption for private tallying combined with Groth16 proofs for cheap EVM verification over BN254 precompiles.

The project starts from a working artifact with a real Groth16 verifier on EVM, a reproduced attack on the naive composition, and benchmarked implementations. The fellowship would fund the formalization of this primitive as an object of programmable institutional design, the strengthening of tally-side integrity beyond the current transcript-auditable path, and the release of a public open-source artifact and academic paper for the Ethereum ecosystem. While private voting is the flagship case, the same research direction is relevant more broadly to encrypted aggregation workflows in which a public system must verify that an encrypted update or contribution is the one authorized by a succinct proof, including emerging work on verifiable aggregation and input verification in federated learning [15, 16].

Objectives

1. **Formalize a governance primitive for private collective decisions.** Define clearly which institutional constraints become feasible through programmability and verifiability in this setting, and map the design into immutable, adjustable, and discretionary components.
2. **Strengthen tally-side integrity in a realistic way.** Starting from the current transcript-auditable construction, design, implement, and formally analyze an aggregate-integrity mechanism that reduces reliance on privileged tally actors and closes the most important gap between accepted encrypted ballots and the published aggregate.
3. **Produce reusable public infrastructure.** Release a public artifact for Ethereum builders, including the core circuit and contract logic, reproducible benchmarks, documentation, and a small reusable developer surface.
4. **Publish an open-access academic output.** Prepare and submit the research to a strong security venue, with an archived artifact and permissive open-source licensing.

Success metrics.

- One submission-ready academic paper delivered during the fellowship period
- One public artifact repository with documentation and reproducible results
- One implemented and evaluated tally-integrity upgrade beyond the current transcript-auditable baseline
- One external round of feedback from Ethereum governance practitioners or adjacent projects

Outcomes

A new design point for Ethereum governance

The practical contribution is not "blockchain voting" in the abstract. It is a governance primitive for higher-assurance private collective decisions on Ethereum: elections, council approvals, committee decisions, sealed governance proposals, and similar workflows where participants need confidentiality without giving up public verifiability. The project targets institutional and organizational settings, not national political elections [8].

The main institutional outcome is that one critical governance constraint can be moved from privileged trust into verifiable code: the decision recorded on-chain must be the same decision authorized by the proof. In the current design, that constraint becomes immutable at acceptance time rather than being left to an operator or coordinator to enforce correctly off-chain.

Auditable and programmable rule enforcement

This project speaks directly to the RFP's emphasis on predictable, enforceable, and selectively controllable governance structures. In this architecture:

- **Immutable:** the acceptance relation between the encrypted object and the proof that authorizes it
- **Adjustable:** governance parameters such as eligibility roots, thresholds, election timing, and application-level workflow policy
- **Discretionary:** optional modules above the cryptographic core, such as ballot-validity policies or institution-specific operational procedures

The result is not only privacy, but a more legible allocation of control: reviewers, auditors, and governance designers can see which rules are hard-coded and which remain intentionally flexible.

Benefit to the wider Ethereum ecosystem

Ethereum has strong governance tooling, but privacy-preserving governance still often relies on specialized workflow actors, off-chain coordination, or design tradeoffs that are difficult to reuse. This project benefits the ecosystem in three ways:

- it turns a currently under-specified mixed-domain composition problem into a documented and reusable design pattern;
- it gives Ethereum builders a public artifact and benchmark baseline for a governance problem that sits directly at the intersection of encryption, succinct proofs, and EVM constraints shaped by BN254 precompiles [1, 2];

- it translates a narrow cryptographic contribution into institutional design language that can be understood by non-cryptographers working on governance products and frameworks.

Grant Scope

Current starting point

This proposal does not start from a blank slate. I already have a working research artifact centered on the acceptance layer of the problem: a Circom circuit, a Solidity contract using a real Groth16 verifier on EVM, a reproduced attack on the naive composition, passing tests, and measured benchmarks at realistic parameters. I also have a paper draft and a detailed internal map of claims, evidence, and open gaps.

This existing base reduces execution risk, but the work is not already complete. The current artifact is strongest on acceptance-side integrity and on public auditability of the tally transcript. The fellowship would fund the next substantive step: turning that base into a clearer governance contribution with stronger tally-side guarantees, public packaging, and a publication-ready research output.

Research work funded by the fellowship

Work Package 1: Formal model and governance framing. I will refine the current technical model into a governance-oriented formal description of the primitive, including:

- the mixed-domain threat model;
- the mapping between cryptographic guarantees and institutional constraints;
- a clear immutable / adjustable / discretionary decomposition; and
- a comparison with nearby governance architectures already used in the Ethereum ecosystem.

Work Package 2: Aggregate-integrity mechanism. I will design, implement, and evaluate a mechanism that strengthens the current tally path beyond mismatch detectability. The exact construction will be chosen only if it meets four conditions:

- it yields a precise formal claim;
- it has clear implementation semantics;
- it can be tested and benchmarked convincingly;
- it does not distort the project away from its governance-primitive core.

Work Package 3: Public artifact and ecosystem packaging. I will release a public repository centered on the reusable artifact rather than the full private lab history. This will include:

- the core circuit and contract logic;
- benchmark data and regeneration instructions;
- threat-model and limitation notes;
- developer-facing documentation; and
- a small reusable package or integration surface for Ethereum builders.

Work Package 4: Academic output and dissemination. I will write and submit the paper, archive the artifact, and produce a concise non-technical explanation for Ethereum governance practitioners.

Expected outputs

- One open-access research paper submitted to a strong security venue
- One public open-source artifact repository under a permissive license
- One archived artifact snapshot, for example Zenodo
- One concise practitioner-facing technical explainer for the Ethereum governance ecosystem

Related Work

This project sits between two literatures that rarely meet cleanly in Ethereum governance practice.

On the governance side, Ethereum already has mature systems for collective decision-making. Snapshot and Snapshot X show the importance of usable and programmable governance. Aragon shows the importance of modular organizational design and permission structures. MACI shows that private on-chain governance is valuable and feasible, but its audited architecture still organizes tallying and proof generation around a coordinator role. Open Vote Network shows a different design point through self-tallying voting, avoiding the threshold-encrypted aggregation path altogether. Helios and the follow-up critique literature remain important reminders that protocol-level and composition-level gaps, not only broken primitives, can undermine voting systems [3, 4, 5, 6, 7].

On the cryptographic side, same-domain composition is much better understood. LegoSNARK provides a powerful modular composition framework inside a proof-friendly domain. SAVER and newer encryption-friendly proof systems such as Tangram illustrate how much easier composition becomes when encrypted objects and proof statements remain within compatible algebraic structures. Recent work on Paillier proofs, including efficient zero-knowledge arguments and direct range proofs, shows that Paillier-side reasoning is possible, but also that it is not free in practice. Modern threshold Paillier systems such as Tiresias show that Paillier remains a serious primitive rather than only a historical one [9, 10, 11, 12, 13, 14].

This broader relevance is not unique to voting. In federated learning, secure aggregation, verifiable aggregation, and input verification are active problems, which reinforces the point that encrypted aggregation plus public or third-party-checkable correctness is an important systems pattern in its own right [15, 16]. I am not proposing a full federated learning deliverable in this fellowship, but the literature helps show that the composition pattern isolated here is not an eccentric one-off artifact of voting.

The specific gap addressed here is narrower and more practical: Ethereum still lacks a clean, reusable primitive for the case where a raw encrypted governance object in an RSA-domain representation must be authorized by a succinct BN254-friendly proof under public bulletin-board semantics. That is the gap I aim to formalize and close. Existing systems either remain in same-domain cryptographic settings, avoid threshold-encrypted aggregation altogether, or accept different workflow assumptions. The project therefore does not claim that encryption, succinct proofs, or private governance are individually missing; it claims that their mixed-domain interface on Ethereum remains under-specified and under-packaged.

About the Applicant

I am a PhD student in Computer Science at Sapienza University of Rome, where I work under the supervision of Alessandro Mei. My broader PhD work is in security, large-scale systems, and open protocols, and I am currently a Visiting Researcher at the University of Zurich. I have published at *USENIX Security* and *IEEE Access*, and I have spent the last years building public infrastructure as well as doing academic research.

I am well suited to this fellowship because this project requires exactly the combination I already have: security-research discipline, real artifact engineering, and practical Ethereum experience. I founded and maintain *BigBrotr*, an OpenSats-funded observability platform for the Nostr protocol, which demonstrates my ability to deliver funded public research infrastructure. I have also built multiple EVM projects, including MeltyFi, Nostreum, and the earlier on-chain election-integrity prototype from which this research direction emerged. This means I am not approaching Ethereum governance as an external observer; I am already an active builder in the ecosystem.

Relevant links

- Personal website: vincenzo.imperati.dev
- Sapienza researcher profile: research.uniroma1.it
- GitHub: github.com/VincenzoImp
- USENIX Security paper: The Conspiracy Money Machine
- IEEE Access paper: Hard to See, Harder to Block
- Project repository link: github.com/VincenzoImp/cross-group-binding
- ORCID: 0009-0001-9437-1384
- Google Scholar: scholar.google.com/citations?user=V4XgGVcAAAAJ
- Supervisor academic profile: Alessandro Mei

Methodology

This is applied cryptography research with a concrete institutional example. The methodological goal is not to simulate a full voting platform; it is to study, strengthen, and package one governance primitive that becomes useful once private decisions and succinct authorization proofs must coexist on Ethereum.

Phase 1: Formalization and architecture comparison

I will start by refining the mixed-domain model from the current research base into a cleaner governance-level statement. This includes:

- specifying the threat model induced by a public bulletin board and cheap on-chain proof verification;
- mapping the primitive into immutable, adjustable, and discretionary governance components;
- comparing it against adjacent design points, especially coordinator-centric, self-tallying, and same-domain approaches.

Phase 2: Design and evaluation of the tally-integrity upgrade

I will then build the main research extension of the fellowship: a stronger aggregate-integrity mechanism. I already know what the present baseline is:

- accepted ballots are bound correctly at the board layer;
- transcript-derived aggregate mismatch is publicly detectable.

The fellowship will fund the step beyond that baseline. I will explore the most credible design path available from the current artifact, implement it in the same narrow stack, and only retain it in the final output if it satisfies the formal and empirical criteria listed above.

Phase 3: Public artifact, documentation, and paper

Once the research line is stable, I will:

- prepare a public artifact repository;
- write a clear threat-model and limitations document;
- archive benchmark outputs and regeneration paths;
- rewrite the paper into venue-ready form; and
- produce a short explainer for practitioners who care about governance design but are not specialists in cryptography.

Scope discipline

The project will remain intentionally scoped. During the fellowship I will not reframe it as:

- a complete national-election system;
- a complete solution to all tally and decryption correctness questions;
- or a three-application general framework spanning voting, federated learning, and MPC simultaneously.

This scope discipline is part of the execution strategy: the goal is to deliver one strong, useful result rather than many half-finished ones.

References

References

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[2] Vitalik Buterin. EIP-197: Precompiled contracts for optimal ate pairing check on the alt_bn128 curve. Ethereum Improvement Proposals, 2017. https://eips.ethereum.org/EIPS/eip-197

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Timeline

Assumed fellowship period:

- June 2026 to May 2027
- approximate total effort: 480 hours
- equivalent internal hourly rate for planning: $50/hour
- total requested fellowship support: $24,000

Milestone 1: Governance model and problem consolidation

Months: 1–3
Hours: 120
Budget: $6,000

Work

- consolidate the formal and governance model
- finalize the immutable / adjustable / discretionary decomposition
- prepare the architecture comparison against adjacent Ethereum governance designs

Deliverables

- stable problem statement
- governance-comparison section
- updated research plan for the tally-integrity extension

Milestone 2: Aggregate-integrity mechanism

Months: 3–6
Hours: 160
Budget: $8,000

Work

- design the tally-integrity upgrade
- implement the mechanism in the existing artifact stack
- add tests and measure cost
- formalize the corresponding claim if the mechanism is strong enough

Deliverables

- implemented and benchmarked tally-integrity upgrade
- updated threat model and limitation notes
- decision on final paper scope

Milestone 3: Public artifact release

Months: 6–8
Hours: 100
Budget: $5,000

Work

- prepare the public repository
- write documentation and reproduction instructions
- package benchmark data and artifact metadata
- archive a stable snapshot

Deliverables

- public artifact repository
- archived release
- practitioner-facing explainer draft

Milestone 4: Paper submission and dissemination

Months: 8–12
Hours: 100
Budget: $5,000

Work

- complete the paper
- submit to a strong security venue

- collect ecosystem feedback
- publish the public explainer and integration notes

Deliverables

- submitted paper
- final public explainer
- short feedback memo from ecosystem conversations